

## 2008 JC2 [H1] Prelim II Paper 2 (Answers)

### Section A

#### Question 1

(a)

increase temperature leads to more kinetic energy;

more kinetic energy leads to more effective collision between substrate & enzyme molecules;

forming more ES complexes;

optimum temp is at 45°C;

after optimum temperature due to high molecular vibrations causing bonds ( hydrogen bonds ) in enzyme to break;

causing 3D configuration and its complementary active site to lose its shape;

resulting in denaturation of enzyme;

(b)

At low [S]

[S] is the limiting factor;

when [S] is increased, there are more substrate molecules to bind to active sites of enzymes;

thus, higher chances of ES complex formation due to higher chance of effective collision;

At high [S]

enzymes' active sites are occupied thus increasing no. of substrate molecules will not cause a change in rate of reaction;

(c)

optimum temperature depends on how long reaction is exposed to high temperature;

longer assays will have more time for enzyme to denature as it is exposed to high temperature for longer period time;

lead to increasing denaturation with more enzymes denatured due to breaking of hydrogen bonds & loss of tertiary structure & active sites;

(d)

increasing temperature/ increasing enzyme concentration



- (b) wrinkled, green: homozygous recessive, thus give rise to only 1 type of gamete;  
 round, yellow: can be either homozygous dominant or heterozygous;  
 heterozygous gives rise to variety of types of gametes, thus offspring are of different genotypes & phenotypes;
- (c)  $0.47 < \text{critical value of } 7.8$ ;  
 no significant difference between observed & expected results  
 difference is due to chance;

### Question 3

- (a) needed to make ATP from ADP and Pi in phosphorylation;
- (b)  $O_2$  needed for formation of ATP in oxidative phosphorylation;  
 in ETC,  $O_2$  acts as terminal electron acceptor & combines with  $H^+$  to produce water;  
 allows recycling of reduced NAD and FAD for glycolysis & Krebs cycle to continue;
- (c) less  $O_2$  available in medium Z than at Y;  
 due to all  $O_2$  have been used up;
- (d) glucose cannot enter mitochondria as it is too large to enter/no carrier system for it;  
 glucose cannot be metabolized as necessary enzymes not present;

### Question 4

- (a) circular d/s DNA;  
 contains only a few genes (e.g. lac operon);  
 contains selectable markers (e.g. tetracycline genes) conferring resistance to bacteria;  
 contains origin of replication (ori), plasmids are self-replicating;  
 contains restriction sites for specific restriction enzymes to digest DNA;

- (b)** R.E. only recognizes specific base sequences on human DNA (e.g. GAATTC by EcoR1);  
R.E. cuts/digests DNA at specific recognition site within recognition sequence (e.g. G-A site by EcoR1);  
recognition seq & restriction site occurs many times in human DNA but only once in plasmid;  
thus cut human DNA in many places & cut plasmid DNA only once;
- (c)** joining of complementary sticky ends of plasmid DNA & human DNA fragment is a random process;  
some plasmids might not have taken up any human fragment or insulin gene inserted at all linearized plasmid recircularizes;  
some plasmids only take up those human fragment without insulin gene;  
some plasmids might have taken up human DNA fragment with insulin gene;
- (d)** growth of bacteria colonies on selective agar plates allow identification of bacteria that have taken up plasmid containing human DNA or not via replica plating;  
use of velvet surface to transfer bacteria colonies from one selective master plate to replica plate;  
use of agar plate (master) containing ampicillin antibiotic & replica plate containing tetracycline antibiotic;  
in bacteria with plasmid inserted with human DNA, its tetracycline resistance gene will be disrupted as human DNA is inserted into its gene sequence;  
thus rendering bacteria sensitive to tetracycline and no growth observed on selective agar plate;  
bacteria with plasmid (w/o any extra human DNA inserted) will grow on both plates (master plate with ampicillin & selective plate with tetracycline) ;  
no growth of colonies on replica plate but corresponding observable growth on master plate infers that the observable colonies grown on master plate contains bacteria with plasmids with human DNA inserted;
- (e)** use of large scale fermenters to produce large quantities of bacteria in shortest time possible;  
provide necessary nutrients plus optimum conditions for optimum growth of bacteria;  
e.g. optimum temperature, effective carbon source; optimum O<sub>2</sub> conc., etc;  
bacteria will reproduce & multiply in large quantities;  
as cells multiply, plasmids also replicate and inadvertently insulin genes also multiply;

insulin genes are expressed into insulin hormones;

insulin hormone accumulates and extracted;

## **Section B**

### **Question 5**

**(a)**

#### Role of mRNA

1. conveys genetic information from DNA in nucleus to ribosomes in cytoplasm;
2. synthesized in nucleus;
3. by transcription using DNA sense strand as template;
4. transported into cytoplasm through nuclear pores;
5. carries code for amino acid sequence of a polypeptide;
6. in the form of codons/triplet bases;
7. codons specify order in which amino acids are arranged to form a polypeptide;
8. provides binding sites for complementary (anti-codon) amino-acyl-tRNA complexes;
9. transient/exists within cell for only a short time;

#### Role of tRNA

10. carries appropriate / specific amino acid;
11. for translation purpose;
12. has anticodon - complementary to codon on mRNA;
13. allow correct sequencing of amino acids on polypeptide coded for by gene;
14. specific amino acid carried on 3' end (CCA stem);
15. tRNA molecules become attached to specific amino acids through action of amino acyl-tRNA synthetases;
16. specific amino acid carried by tRNA is determined by anti-codon of tRNA;

**(b)**

1. mutations in cancer critical genes;
2. leading to uncontrolled/excessive cell division;
3. cancer is a multi-step process and requires multiple / accumulation of mutations;

Proto-oncogenes

4. gain-of-function mutations; only one copy of gene is mutated activating oncogene;
5. mutation to oncogene results in uncontrollable cell proliferation;
6. mutations in promoters or control elements lead to up-regulation of proto-oncogene expression;
7. mutations within proto-oncogene coding region;
8. result in a constitutively active gene product / gene product that binds to its substrate with increased affinity;

Tumor suppressor genes

9. loss-of function mutations;
10. both copies of gene need to be mutated;
11. loss of tumor suppressor genes allows loss of control and inappropriate cell cycle progression;
12. mutations in promoters or control elements; lead to down-regulation of tumor suppressor gene expression;
13. mutations within tumor suppressor gene coding region;
14. result in a constitutively inactive gene product / gene product that binds to its substrate with decreased affinity / or a non-functional gene product due to loss of protein structure;

(c)

1. occurs in nucleus of eukaryotic cell;

#### Capping

2. 5' end of pre-mRNA transcript is capped by addition of a guanylyl residue;
3. cap structure is methylated at the 7-position of the G residue;
4. forming 5' end of pre-mRNA with methylated guanine cap;

#### Polyadenylation

5. adding of poly-A tail to 3' end of pre-mRNA;
6. poly A addition signal is at the consensus sequence of AAUAAA upstream of addition site;
7. prevent enzymatic degradation in cytoplasm;

#### RNA Splicing

8. removing of introns (non-coding regions) of pre-mRNA;
9. by spliceosomes at specific splice sites;
10. splicing/joining of exons (coding regions) of pre-mRNA;

#### Alternative splicing

11. takes place to include different exons from a pre-mRNA;
12. increases coding capacity of genome.
13. allows variations of proteins to be produced, increasing protein diversity;
14. mature mRNA formed leaving nucleus through nuclear pores into cytoplasm;

## Question 6

(a)

1. **DNA template** - sample DNA that contains target sequence;
2. **DNA polymerase** - a type of enzyme that synthesizes new strands of DNA complementary to target sequence, *Taq* DNA polymerase;  
they can generate new strands of DNA using a DNA template and primers, and they are heat resistant;
3. **primers** - short pieces of single-stranded DNA that are complementary to target sequence; polymerase begins synthesizing new DNA from end of primer;
4. **nucleotides (dNTPs or deoxynucleotide triphosphates)** - single units of bases A, T, G, and C, which are essentially "building blocks" for new DNA strands;
5. 3 major steps in a PCR, which are repeated for 30 or 40 cycles; done in automated cycler, which can heat & cool tubes with reaction mixture in a very short time;
6. denaturation at 94 °C:  
during denaturation, d/s DNA separates to form 2 single stranded DNA, all enzymatic reactions stop;
7. annealing at 54 °C:  
ionic bonds are constantly formed & broken between single stranded primer & single stranded template; more stable bonds last a little bit longer (primers that fit exactly) and on that little piece of double stranded DNA (template & primer), polymerase can attach & starts copying template. Once there are a few bases built in, ionic bond is so strong between template & primer, that it does not break anymore;
8. extension at 72 °C:  
bases (complementary to template) are coupled to primer on 3' end (polymerase adds dNTP's from 5' to 3', reading template from 3' to 5' side, bases are added complementary to template);



**(b)**

1. southern blotting using master plate with transformants, each with vector carrying different fragments of chromosomal DNA → nitrocellulose filter laid gently on master plate to pick up some bacteria from each colony;
2. filter is then treated with detergent to lyse bacteria & then NaOH added to denature its DNA → denatured ssDNA will adhere to filter;
3. filter is then submerged in a solution containing radiolabelled DNA probe which is complementary to gene of interest → DNA probe will hybridise to DNA of interest;
4. filter is then washed to remove unbound probes & then placed next to X-ray film → colonies with hybridised probes will show up as a dark spot on film in corresponding location;
5. to check if an endangered species has been bred in captivity do southern blot using DNA of test organism & other organisms in captivity to test for its parents;
6. if organism has been in captivity then it should have same bands as father & mother that has been kept in captivity as well;  
(comparing bands)

**(c)**

Genomic Library

1. genomic library refers to a set of DNA clones collectively containing entire genome of an organism;
2. restriction enzymes are added to genome of interested specimen e.g. rat genome;
3. shotgun method used to generate restriction fragments;
4. all restriction fragments yield would be ligated into vectors & transformed into bacterial cells;
5. transformed cells will be plated & selected on agar plates with selectable markers;

cDNA library

6. cDNA library is a set of DNA clone containing cDNA that was being reverse transcribed from mRNA;
7. mRNA with 5' guanine cap & 3' poly(A) tail used as starting material;
8. poly(T) oligonucleotides added to act as primers for reverse transcriptase;
9. elongation can proceed by 3 ways:
  - a. reverse transcriptase synthesise hairpin loop;
  - b. addition of oligo(dC) bases by terminal transferases;
  - c. degraded RNA fragments act as primers,